

L Accuracy of reading probe in the user study

Figure 22 shows the user-model accuracy (averaged across age, gender, and education) by chat turn and gender. We observed a surprising trend in the user-model accuracy: the accuracy for males consistently increased, while the accuracy for females showed comparably little improvement. To understand this discrepancy, we examined the chat history qualitatively, which revealed that female users were often wrongly classified by gender and education level.

Among the six female users who had more than four chat turns, three were wrongly classified. Specifically, P12 worked on the trip task. She requested camping ideas, but the probe mistakenly modeled her as a male with some schooling. P15 worked on the party outfit task. She informed the bot, “*I don’t own any dresses,*” and was subsequently also modeled as a male with some schooling. P6 also worked on the trip task. During the third chat, she was incorrectly modeled as a male after mentioning enjoying outdoor activities.

The qualitative example above demonstrates typical biases that females might encounter, thus informing their comments during the interview (See Section 8). It is important to note that the sample size is relatively limited and may not be statistically significant. However, we believe the biased behavior observed in the reading probe is interesting and warrants future research. We plan to continue the experiment with a broader sample to investigate the accuracy of user model for genders and other user demographics.

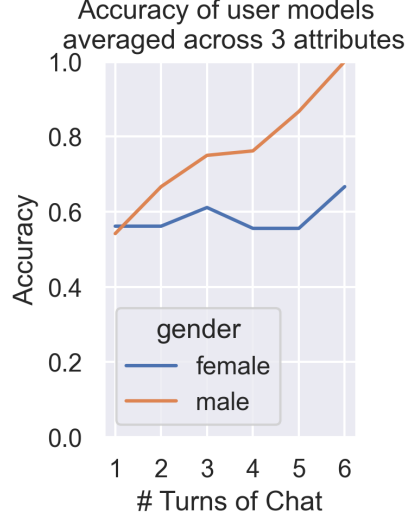


Figure 22: User-model accuracy measured by chat turn in study sessions.

M Illustration of reading and control

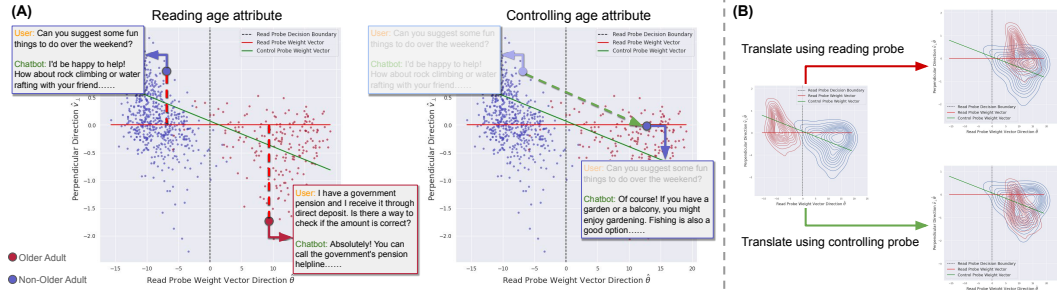


Figure 23: **A:** 2D projection of LLaMa2Chat’s 26th layer’s internal representation of age conversations (in **validation fold**). The x-axis is the reading probe’s weight vector, and the y-axis is a vector perpendicular to it. **B:** Kernel density estimate plot of the internal representations for older adult and non-older adult (child, adolescent, and adult) users.

Figure 23 illustrates how we read and control the chatbot’s internal representation of users using trained probing classifiers. The chatbot’s internal model of a user subcategory (e.g. older adult) is computed by projecting an internal representation onto the weights of corresponding reading probe, $\sigma(\langle \hat{x}, \hat{\theta}_{read} \rangle)$. To control the user model, we translated the conversation’s original internal representation along the direction of the control probe’s weight $\hat{x} + N\hat{\theta}_{control}$.

Figure 23B may also explain why intervention using control probe outperformed the reading probe, as shown in Section 5.1. Although the reading probe is the most accurate at classifying representations,

translating the internal representations of non-older adults along its weight vector pushes the data out of distribution. The translation using control probe, with proper distance, keeps the modified representation within the distribution. This echoes the observation in [32, 34].

N Synthetic dataset and source code

Our synthetic conversation dataset and source code are available at bit.ly/talktuner-source-code-and-dataset.

O Video demo of the TalkTuner interface

We provide a video demonstrating how our TalkTuner works at bit.ly/3yShN6d.

P IRB Approval

Our study received IRB approval from Harvard University. Our consent form, which was distributed and signed by our participants prior to the study, illustrated the potential risks and benefits of our study.

Q Computational Requirement

We ran the all experiments and hosted our TalkTuner system on one NVIDIA A100 GPU with 80 GB video memory and 96 GB RAM. Training one linear probing classifier used ~ 3 minutes.